CSE 516 — PROBABILISTIC GRAPHICAL MODELS / APPLIED AI, GROUP G9

# CausalLOB: What Does a Sparse Temporal Graph Add to Limit-Order-Book Inference?

Final research report — Milestone 4 | Financial market microstructure

**Abstract.** We ask whether a sparse temporal graph adds useful information to limit-order-book (LOB) inference beyond autoregression. From 394,337 FI-2010 records, we construct six market-state variables, screen lagged relationships with LPCMCI, and fit a dynamic Bayesian network with 39 frozen parent slots. On three previously inspected days, return reconstruction improves over AR(5) (RMSE 0.824720 versus 0.833930); dense VAR(5) performs slightly better (0.822980). A DeepLOB-style model achieves lower matched forecasting log-loss at all five horizons. During imposed gaps in observation history, even a fixed training-frequency prior outperforms the DBN in every tested setting. We examine these limits alongside exact missing-state inference, calibration, day-level variability and a separately reproduced 40-replication synthetic mechanism-change experiment. Original checkpoints and saved forecast probabilities were audited. The results support a limited role for transparent probabilistic state inference, but neither causal identification nor forecasting dominance.

## 1 Introduction and research contract

Across the four milestones we ask the same question: which lagged order-book states help explain short-horizon midpoint movements, and can those relationships support inference when observations are missing? Depth and spread are persistent; imbalance can reverse quickly; returns are noisy. A directional forecast alone cannot answer what happens when a state measurement is unavailable.

M1 established the FI-2010 data pipeline, six state variables and LPCMCI plan [5]. M2 froze the 39-slot graph and fitted its transition and outcome distributions [6]. M3 compared transition, prediction and missing-state performance with alternative models [7]. Here we consolidate those results and audit the original frozen checkpoints. The graph and market-forecast models were not changed for M4; archived market results were not recomputed by full retraining.

The test is deliberately narrow: do nine selected cross-variable lags add useful information beyond each target's five own-history lags? The contribution lies in explicit, inspectable conditional distributions, not a new learning algorithm.

## 2 Prior work and SOTA position

**Structure learning.** Gerhardus and Runge developed LPCMCI for autocorrelated series with possible hidden common causes [2]; it motivates our screening step, not the subsequent inference model. PCMCI+ offers a related comparison under different assumptions [3]. An edge retained here describes an observational temporal relationship. It does not establish what would happen if a trader intervened on a quote or queue.

**Forecasting.** DeepLOB predicts direction from raw order-book windows [4]. We use a DeepLOB-style implementation as a forecasting reference, not a reproduction of the published benchmark. Its $100 \times 40$ input and model capacity differ substantially from the six-state DBN; their performance gap cannot be attributed to graph learning alone. FI-2010 supplies the book records and three-class publisher labels [1]. The original message-level label-generation process was not independently reproduced.

**Table 1:** Chronology and frozen model contract.

| | |
|---|---|
| Structure search | LPCMCI on Days 1–5, maximum five representation-step lags. |
| Graph validation | Days 6–7; 59 initial links, 16 cross-variable source–target families screened, five families retained. |
| Parameter fitting | Days 1–7, Ridge $\alpha = 1$; graph frozen beforehand. |
| Retrospective evaluation | Days 8–10 (already inspected in development); 139,212 eligible transition rows. |
| Matched neural forecast | DeepLOB-style windows of shape $100 \times 40$, train Days 1–5, select Days 6–7, evaluate identical rows by horizon. |
| Inferential unit | Trading day; just three retrospective evaluation days. |

## 3 Data and experimental protocol

The canonical FI-2010 NoAuction Decimal Precision archive holds 394,337 representations, 40 raw price/volume columns and five publisher labels. Each day contains five concatenated anonymous securities, yielding 50 day-by-security sequences. Their boundaries stop lag and window leakage; they are not 50 independent replications. Horizons $h \in \{10, 20, 30, 50, 100\}$ denote *underlying publisher-message horizons*, not physical seconds or future representation indices.

The independently supplied `fi2010_canonical_raw40_labels.npz` passed a SHA-256 audit, the $394337 \times 40$ and $394337 \times 5$ shape checks, the publisher label-set check $\{1, 2, 3\}$ and a per-day count of five segment starts. These checks establish the archive's internal consistency, not the publisher's labeling procedure or a new training run. The original Step 2 feature checkpoints are separately preserved (Appendix C).

## 4 Graph representation and learning

### 4.1 Nodes, edges and factorization

At representation step $t$, define

$$X_t = (s_t, d_t, q_t^{(1)}, q_t^{(2:5)}, o_t, r_t)^\top \in \mathbb{R}^6. \tag{1}$$

Here $s$ is log relative spread; $d$ is log aggregate L1–L5 depth; $q^{(1)}$ is the L1 queue imbalance; $q^{(2:5)}$ is additional deeper imbalance; $o$ is a depth-normalized *quote-change OFI proxy*; and $r$ is one-step log midpoint return. This

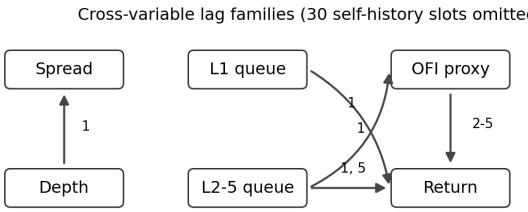



**Figure 1:** Frozen lagged cross-variable families; arrow labels denote representation-step lags. The 30 self-history slots are omitted.

OFI variable is not executed-order event flow. Twenty-step volatility is a regime diagnostic, not an extra primary node.

Each component retains lags 1–5 of itself (30 links). The five retained cross-variable families contribute another nine lagged slots: L1 imbalance → return (lag 1); L2–L5 imbalance → return (1,5) and OFI (1); depth → spread (1); OFI → return (2–5). The graph contains 39 of 180 possible directed lag slots (21.7%), including nine of 150 possible cross-variable slots (6%). These figures measure sparsity, not recovery of a known causal graph.

For every target $j$, the trained transition CPD is

$$X_{j,t} = c_j + \sum_{(i,\ell)\in \mathrm{Pa}(j)} \beta_{ji\ell} X_{i,t-\ell} + \epsilon_{j,t}, \quad \epsilon_{j,t} \sim \mathcal{N}(0, \sigma_j^2). \tag{2}$$

It factors as

$$p(x_{1:T}) = p(x_{1:5}) \prod_{t=6}^{T} \prod_{j=1}^{6} p(x_{j,t} \mid x_{\mathrm{Pa}(j,t)}). \tag{3}$$

The main innovation covariance $Q$ is diagonal, which explicitly assumes independent same-step innovations conditional on their selected lagged parents. A correlated-$Q$ sensitivity therefore changes the graphical assumptions; it is not the frozen primary DBN.

### 4.2 Parameter versus structure learning

LPCMCI supplies candidate structure, and the Days 6–7 held-out-from-discovery checks compare family value against equal-history AR(5), nonlinear prediction, lead placebos and phase-shift references before graph freeze. Five families survived screening. A post-selection random-set diagnostic gave $p \approx 0.119$; this exploratory statistic does not establish a causal graph [6]. Conditional on the frozen parent mask, separate Ridge regressions estimate 39 slopes, six intercepts and six diagonal innovation variances: 51 reported transition scalars, versus 42 for own-history AR(5) and 192 for a fully dense lag-five six-state transition with analogous variances. Five ordered-probit outcome heads are then fitted on Days 1–7. Structure is not retuned on the later evaluation days.

## 5 Probabilistic inference

### 5.1 Exact missing-state query

For current observed components $o$, missing components $m$ and five-lag history $H_t$, the query is $p(X_{t,m} \mid X_{t,o}, H_t)$. From $X_t \mid H_t \sim \mathcal{N}(\mu_t, Q)$, Gaussian conditioning yields

$$\mu_{m|o} = \mu_m + Q_{mo} Q_{oo}^{-1} (x_o - \mu_o), \tag{4}$$

$$Q_{m|o} = Q_{mm} - Q_{mo} Q_{oo}^{-1} Q_{om}. \tag{5}$$

**Table 2:** Pooled retrospective one-step transition metrics (all lower is better). The random row is a median of matched alternatives.

| Model | State RMSE | Joint NLL | Return RMSE | Return NLL |
|---|---|---|---|---|
| AR(5) | .588347 | 3.725577 | .833930 | 1.265020 |
| Sparse-39 | .586137 | 3.712316 | .824720 | 1.251983 |
| Dense VAR(5) | **.585171** | **3.705322** | **.822980** | **1.249820** |
| 100-graph median | .586529 | 3.714656 | .826415 | 1.254092 |

Under the fitted Gaussian assumptions this calculation is exact. Monte Carlo sampling is used only to check the numerical approximation. Under diagonal $Q$, an observed current node cannot directly update a missing node's same-time innovation; only its earlier lagged predictors help. Missing *consecutive* observations require propagation over an augmented lag state, not simply filling each independently.

### 5.2 Probabilities for future outcomes

Each fitted ordered-probit head has latent score $U_h = \beta_h^\top X + \eta$, $\eta \sim \mathcal{N}(0,1)$ and ordered cutoffs $a_h < b_h$. Integrating over the Gaussian conditional state posterior $X \mid E \sim \mathcal{N}(\bar{\mu}, S)$ gives

$$U_h \mid E \sim \mathcal{N}(\beta_h^\top \bar{\mu},\ 1 + \beta_h^\top S \beta_h). \tag{6}$$

Down, Stationary and Up probabilities are the associated consecutive standard-normal CDF differences (Appendix A). They sum to one by construction, but normalized probabilities need not be empirically calibrated.

## 6 Baselines, metrics and fairness

AR(5) Ridge uses each target's own lags and the same transition training rows and standardization; it isolates the extra cross-lag contribution. Dense VAR(5) uses all six variables' five lags. One hundred target-and-lag-count-matched source-randomized graphs test whether chosen source identities matter; they share the same market history and do not represent independent replications. A static reference tests the value of temporal persistence itself. A Days 1–7 class-frequency prior provides a demanding baseline for forecast probability quality. DeepLOB is scored on matched row IDs and publisher labels, but its larger raw-book input makes it a forecasting reference, not an equal-capacity structure ablation.

The selected KPIs cover: *representation* (nodes, parent slots, sparsity); *learning* (state and return RMSE, Gaussian joint/return negative log-likelihood); *inference* (missing-state RMSE, joint NLL, marginal 90% coverage error, error versus runtime); and *prediction* (three-class log-loss, multiclass Brier, confidence expected calibration error, balanced accuracy). Brier is $n^{-1} \sum_{i,k} (p_{ik} - \mathbf{1}[y_i = k])^2$; log-loss is $-n^{-1} \sum_i \log p_{i,y_i}$. Lower scores are better except balanced accuracy and coverage proximity to nominal. Since FI-2010 has no validated ground-truth causal graph, real-data SHD/SID would be invented and are not reported.

## 7 Results on the frozen FI-2010 study

### 7.1 Transition fit: gains are small and conditional

Sparse-39 reduces return RMSE by 0.009210 at printed precision, or approximately 1.10% relative to AR(5); pooled state RMSE improves by only 0.002210. The exact source comparison prints 0.009211 and 0.002209

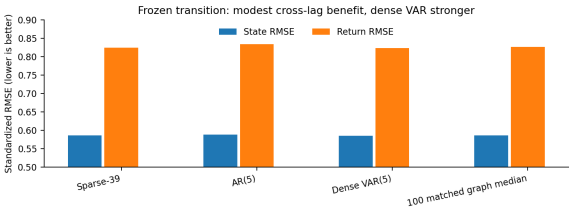



**Figure 2:** Transition reconstruction from archived M3 scores. VAR(5) wins the displayed pooled errors.

**Table 3:** One-step ablations: AR(5) minus Sparse-39. A positive gain favors the sparse model. Coverage column is improvement in absolute marginal 90% coverage error.

| Missing current components | RMSE gain | Joint NLL gain | Cov. error gain |
|---|---|---|---|
| L1 imbalance | .000000 | .000000 | .000000 |
| OFI proxy | +.000332 | +.000661 | −.000115 |
| Both imbalances | .000000 | .000000 | .000000 |
| Liquidity | −.000207 | −.000438 | +.000672 |
| Return + OFI | +.004752 | +.013698 | −.005984 |
| History only | +.002209 | +.013261 | −.001771 |

**Table 4:** Five horizons on strictly matched rows (archived Step13/15 output). LL: multiclass log loss; Brier: multiclass Brier. All are lower-is-better.

| $h$ | $n$ | LL Deep | LL DBN | LL Prior | Brier Deep | Brier DBN |
|---|---|---|---|---|---|---|
| 10 | 137952 | **.4546** | .4984 | .8292 | .2426 | **.2368** |
| 20 | 137802 | **.6459** | .7142 | .9578 | **.3614** | .3864 |
| 30 | 137652 | **.6015** | .9368 | 1.0302 | **.3332** | .5405 |
| 50 | 137352 | **.5606** | 1.0711 | 1.1096 | **.3105** | .6433 |
| 100 | 136602 | **.5695** | 1.1477 | 1.1614 | **.3158** | .6906 |

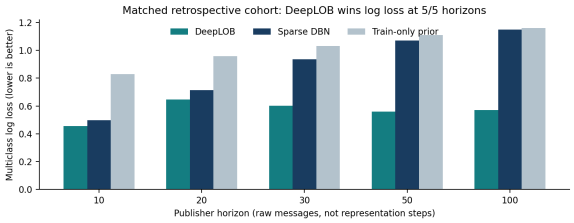



**Figure 3:** Matched probabilistic forecast log-loss. DeepLOB has smaller log-loss at every publisher horizon.

before rounding. Dense VAR(5) does better on every pooled transition metric. Sparse-39 beats 85, 90, 93 and 95 of 100 matched random-source graphs on state RMSE, joint NLL, return RMSE and return NLL, respectively. The graph was selected using the same underlying dataset, so the ranks are descriptive rather than selection-adjusted significance tests. The history-only-versus-static RMSE advantage (0.360443) is much larger than the sparse-versus-AR gain: temporal persistence supplies most of the signal.

### 7.2 Missing-state inference and ablations

The largest mask-specific improvements appear when return and OFI are missing. The liquidity mask instead favors AR(5), and two masks give zero incremental gain. In the return-plus-OFI case, RMSE and joint NLL improve but absolute marginal coverage error *worsens* by 0.005984. Better point estimates do not necessarily produce better probability intervals. Which masks benefit depends on the selected parents and the diagonal-$Q$ assumption.

### 7.3 Forecast probability quality against DeepLOB

At $h = 10$, the DBN's balanced accuracy is 0.727736 versus DeepLOB's 0.676273, and its Brier score is 0.236803 versus 0.242605, even though its log-loss is worse (0.498405 versus 0.454644). The short-horizon result depends on the scoring objective: log-loss weighs badly assigned probability to the realized class more harshly. At $h = 20, 30, 50, 100$, the neural comparator has lower log-loss and Brier. The retained six-state graph is not the better general purpose forecaster. Class-prior Brier at horizons 10/20/30/50/100 is 0.471867/0.565968/0.618830/0.673864/0.700842; all five full-evidence DBN Brier values are lower than this prior, but this does not transfer to consecutive gaps.

### 7.4 Calibration is a separate assessment

The DBN beats the training-frequency prior on confidence ECE only at $h = 10$, despite its lower log-loss at all five horizons in this cohort. ECE depends on binning and measures confidence calibration rather than full classwise reliability. Using the newly supplied original Step6 probability arrays, we reconstructed all five 15-bin reliability tables and verified that their recalculated ECE values agree with the archived Step6 outputs at the printed precision. Figure 5 displays two illustrative horizons. These are retrospective full-evidence DBN reliability diagrams, *not* recalculated DeepLOB-matched reliability curves.

## 8 Robustness and failure analysis

### 8.1 Consecutive observation gaps

The Step9 augmented filter used 697 spaced endpoints (277/260/160 on Days 8/9/10). It tested two missing-evidence patterns, gaps of 1, 5 or 20 representation steps, and forecast horizons 10 and 20. Across all 12 settings the Days 1–7 class-frequency prior had lower log-loss. The prior-minus-DBN differences range from −0.183964 to −0.078269 (Table 8); a negative value favors the prior. These were imposed gaps, not exchange outages. As uncertain states are propagated forward, the forecasting head receives inputs farther from its training distribution. The failure against a constant prior matters: good one-step reconstruction does not imply reliable predictions after missing history.

### 8.2 What robustness is and is not covered

The frozen notebook includes comparisons across masked components, alternative graph sources, three retrospective day blocks, Gaussian innovation-covariance sensitivities and deliberate consecutive gaps. It does not include a full FI-2010 sample-size sweep, known-ground-truth SHD benchmark or measured market-noise perturbation study. Those experiments remain open. A deployed model would also need a tested rule for falling back to simpler forecasts when recent observations are missing.

## 9 Statistical variation and controlled simulation

### 9.1 Market data: three descriptive day blocks

Observations overlap heavily within each day, and the five security segments are not independent trading-day replications. Days 8–10 were also inspected during model development. A row-level paired $t$-test would understate uncertainty; three retrospective days cannot support a

**Table 5:** Confidence ECE on the *Step6 full-observation cohort*, which differs from the matched DeepLOB cohort. Lower is better.

| $h$ | DBN ECE | Prior ECE | DBN LL | Prior LL |
|---|---|---|---|---|
| 10 | **.088780** | .101577 | .499358 | .830344 |
| 20 | .135478 | **.126579** | .715190 | .958605 |
| 30 | .195836 | **.143100** | .937313 | 1.030661 |
| 50 | .103171 | **.068967** | 1.070878 | 1.109434 |
| 100 | .086855 | **.063303** | 1.146834 | 1.160453 |

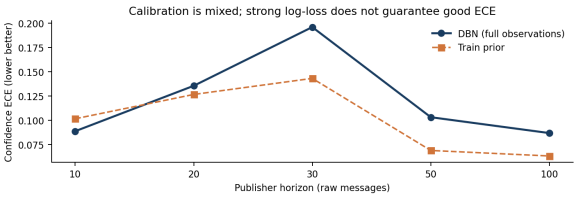



**Figure 4:** Full-observation confidence calibration. ECE ranks the models differently from log-loss; it cannot be pooled with the DeepLOB-matched scores.

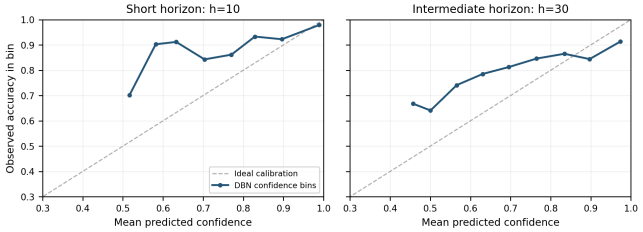



**Figure 5:** Original checkpoint-based DBN confidence reliability for $h = 10$ and $h = 30$; 15 fixed-width bins with nonempty groups. Source: saved Step6 row probabilities, not generated predictions.




**Figure 6:** All 12 archived artificial consecutive-gap comparisons favor the training-only prior.

credible confirmatory confidence interval or selection-adjusted $p$-value. The displayed means and SDs come from saved per-day scores, not a fresh DeepLOB inference run. The 100 graph alternatives share the same market history. We therefore treat all market comparisons as descriptive.

### 9.2 Separately reproduced known-mechanism simulation

To test mechanism dependence without claiming new market data, Step16 constructs six-dimensional artificial Gaussian series. In training, $x_{5,t} = 0.45x_{5,t-1} + 0.80x_{0,t-1} + \varepsilon_t$; test data either preserve $+0.80$ or deliberately flip that cross-coefficient to $-0.80$. Across 40 independent RNG replications (seed 5162026), each model fits 800 training targets and predicts 300 targets under each regime, without test-time fitting. The comparators are self-history AR(5), a known one-edge model, a deliberately wrong edge and dense VAR(5). The test isolates the consequence of relying on a relationship that later changes sign. It does not evaluate LPCMCI recovery or market generalization.

The known-edge model beats AR(5) in 40/40 stable replications, but in 0/40 after the deliberate flip. For the paired RMSE difference $\Delta = \text{RMSE}_{AR5} - \text{RMSE}_{known}$, the stable regime has $\bar{\Delta} = +0.24443$ (95% $t$ CI $[0.23575, 0.25312]$); the flipped regime has $\bar{\Delta} = -0.64228$ (95% $t$ CI $[-0.66580, -0.61876]$). Paired tests on the 40 independent simulated replications give $p < 10^{-30}$ for both comparisons. The intervals and tests apply only to the simulated process; they do not supply market-data confidence intervals. The simulator was independently rerun from its frozen Step16 equations, and all 320 model-by-regime-by-replication scores matched the archived output to numerical tolerance.

### 9.3 Numerical inference error versus computation

The M3 supplementary notebook also evaluates exact six-dimensional Gaussian posterior means against Monte Carlo approximations using a separate artificial numerical fixture. Across 50 RNG replicates, posterior-mean RMSE is $0.1494 \pm 0.0730$ at 32 draws, $0.0423 \pm 0.0218$ at 512 and $0.00825 \pm 0.00561$ at 8,192. A reported 95% interval for the simulated mean at 8,192 draws is $[0.00666, 0.00984]$. This confirms expected numerical convergence within that fixture. Its measured runtime depends on the device and implementation; it is not a fresh latency benchmark for the actual stored DBN. Exact analytic conditioning remains the computational default because the state is only six-dimensional.

## 10 Discussion: what the result means

The nine cross-variable links help in selected reconstruction and missing-state tasks. They do not provide the best overall transition fit: dense VAR(5) performs slightly better. Nor do they deliver the best forecasts: DeepLOB wins matched log-loss, and the training prior wins when history is interrupted. The DBN's advantage is more specific. Its conditional distributions expose the role of each retained lag and permit exact posterior queries under stated independence assumptions.

Relative to LPCMCI [2], our extension is the fitted DBN and its missing-state queries; we do not reproduce LPCMCI's graph-recovery benchmark or identify a market causal DAG. Relative to DeepLOB [4], the smaller state representation sacrifices forecasting power for transparency, although the unequal inputs preclude a clean architectural comparison. CausalLOB is best understood as a measured study of interpretable temporal inference rather than a new forecasting benchmark.

## 11 Limitations and next experiments

The study uses only ten observational trading days, three of which had already been examined during development. Anonymous securities and overlapping windows further limit statistical independence. We cannot independently recreate the publisher's message-level labels. Diagonal Gaussian innovations and ordered-probit heads may miss shared shocks or changing regimes, while the imposed missingness patterns may not resemble real outages. DeepLOB uses richer inputs and greater capacity, and no after-cost trading, fills or latency were tested. Finally, the known-mechanism simulation is not evidence of recovering market structure.

**Table 6:** DeepLOB log-loss advantage over DBN on Days 8–10: unweighted day mean ± sample SD, recomputed from the original archived Step12 three-day score rows. These are descriptive, not inferential confidence intervals.

| $h$ | Days better | Mean ± SD | Daily range |
|---|---|---|---|
| 10 | 3/3 | .0397 ± .0222 | [.0146,.0571] |
| 20 | 3/3 | .0637 ± .0260 | [.0362,.0879] |
| 30 | 3/3 | .3294 ± .0320 | [.2938,.3556] |
| 50 | 3/3 | .5032 ± .0417 | [.4602,.5434] |
| 100 | 3/3 | .5606 ± .0963 | [.4557,.6450] |

**Table 7:** Synthetic RMSE, 40 replications (mean ± sample SD). Model selection uses no test data.

| Model | Same constructed DGP | Coefficient flip |
|---|---|---|
| AR(5) | .6952 ± .0317 | **.6983** ± .0257 |
| Wrong-edge | .6961 ± .0317 | .6991 ± .0258 |
| Known-edge | **.4508** ± .0189 | 1.3405 ± .0809 |
| Dense VAR(5) | .4581 ± .0200 | 1.3628 ± .1028 |

The next evaluation should pre-register its comparisons before collecting new trading days and instruments. It should preserve message-level label provenance, measure calibration and missing-state coverage, and set a fallback rule before testing it. Blocked uncertainty estimates need more independent trading days. Controlled changes in training size, noise and market regime would then test whether the sparse graph is stable beyond FI-2010.

## 12 Conclusion

CausalLOB combines an explicit temporal graph, separate structure and parameter learning, and exact conditional inference. The nine cross-variable lags modestly improve retrospective return reconstruction over AR(5), but dense VAR(5) fits better overall, DeepLOB forecasts better by log-loss, and the DBN is fragile when recent observations disappear. Its useful result is therefore conditional: a compact, inspectable probabilistic representation that helps on some inference queries, with clearly measured limits.

## References


[1] A. Ntakaris et al., "Benchmark Dataset for Mid-Price Forecasting of Limit Order Book Data with Machine Learning Methods," *Journal of Forecasting*, 37(8), 852–866, 2018.

[2] A. Gerhardus and J. Runge, "High-Recall Causal Discovery for Autocorrelated Time Series with Latent Confounders," *NeurIPS*, 2020.

[3] J. Runge, "Discovering Contemporaneous and Lagged Causal Relations in Autocorrelated Nonlinear Time Series Datasets," *UAI / PMLR 124*, 1388–1397, 2020.

[4] Z. Zhang, S. Zohren and S. Roberts, "DeepLOB: Deep Convolutional Neural Networks for Limit Order Books," *IEEE Transactions on Signal Processing*, 67(11), 3001–3012, 2019.

[5] CausalLOB Group G9, *Milestone 1: Proposal and Scoping*, CSE 516 course submission.

[6] CausalLOB Group G9, *Milestone 2: Technical Foundation for a Dynamic Bayesian Network of LOB States*, CSE 516 course submission.

[7] CausalLOB Group G9, *Milestone 3: Results and Analysis*, CSE 516 course submission.

[8] CausalLOB Group G9, *PGM Project: frozen source notebook, Steps 2–9 and 12–17*, supplied executed notebook; local SHA-256 recorded in Appendix C.

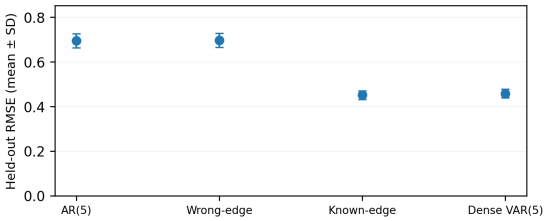




**Figure 7:** Same synthetic DGP: source-informed prediction improves RMSE. Points and error bars are mean and between-replication SD.

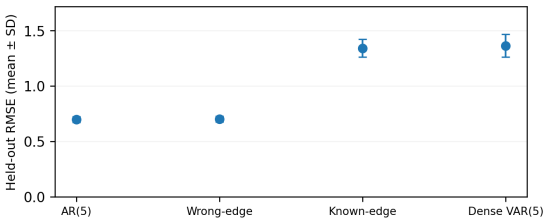



**Figure 8:** Synthetic coefficient flip: the same trained graph becomes worse than AR(5). No real-market shift is asserted.

## A Conditional outcome distribution

Let $X \mid E \sim \mathcal{N}(\bar{\mu}, S)$ and let the ordered probit cutoffs be $a_h < b_h$. Define $v_h = \sqrt{1 + \beta_h^\top S \beta_h}$, $z_{1h} = (a_h - \beta_h^\top \bar{\mu})/v_h$ and $z_{2h} = (b_h - \beta_h^\top \bar{\mu})/v_h$. Then

$$P(\text{Down} \mid E) = \Phi(z_{1h}),$$
$$P(\text{Stationary} \mid E) = \Phi(z_{2h}) - \Phi(z_{1h}),$$
$$P(\text{Up} \mid E) = 1 - \Phi(z_{2h}).$$

For the sparse Gaussian transition, $Q = \text{diag}(\sigma_1^2, \ldots, \sigma_6^2)$, so $Q_{mo} = 0$ under a current-state mask and observed current components cannot alter the missing components' mean beyond the prior given history. This explains why the additional lagged parents, rather than same-step cross-covariance, drive primary one-step mask improvements. Correlated-$Q$ tests change this property and are explicitly sensitivity analyses.

## B Complete gap and evidence scorecards

**Table 8:** All twelve consecutive-gap prior-minus-DBN log-loss differences. Every negative value favors the training-frequency prior.

| Evidence mask | Gap | $h = 10$ | $h = 20$ |
|---|---|---|---|
| Return + OFI | 1 | −.182523 | −.086164 |
| Return + OFI | 5 | −.180089 | −.084771 |
| Return + OFI | 20 | −.180074 | −.084697 |
| History only | 1 | −.181718 | −.078269 |
| History only | 5 | −.183560 | −.086300 |
| History only | 20 | −.183964 | −.089885 |

## C Reproducibility specification and limitations

The final technical archive contains the unchanged frozen notebook, the M3 numerical re-analysis notebook, canonical NPZ, original checkpoint export split into browser-upload-sized parts, CSV tables, figures and three independent validation scripts. The export contains 244 original files (including model weights, saved predictions and stage results), with 26 explicitly required checkpoint paths. It supports checkpoint integrity and score replay;

**Table 9:** What each result can establish. 'Archived' refers to executed notebook outputs, not a market rerun in this deliverable.

| Evidence family | Source and status |
|---|---|
| Canonical shape, day counts, boundaries | Independently checked local NPZ and digest; original Step2 artifacts included. |
| 39-slot graph, Gaussian fits, outcomes | Original frozen Step4/Step6 checkpoints; mask parity and original digests verified. |
| AR/dense/random, missing-state and gap metrics | Original frozen Step7–9 score tables included; main pooled scores and twelve gap contrasts cross-checked. |
| Matched DeepLOB and score replay | Original DeepLOB weights and saved predictions included; archived matched scores cross-checked, no neural-forward rerun. |
| 40-seed mechanism-change experiment | Independently rerun locally from frozen Step16 simulation equations. |
| 50-replicate Gaussian MC convergence | Archived M3 separate synthetic numerical fixture; not the fitted market DBN. |

it does not establish that all models were retrained from scratch.

**Frozen export checks, from `M4_G9/`:** Run the exact commands and dependency versions documented in `Code/README.md`. The principal independent checks are:

```
python Code/src/restore_original_checkpoints.py
python Code/src/verify_frozen_checkpoints.py
python Code/src/rebuild_reliability.py
python Code/src/reproduce_synthetic.py
```

The archive check validates the canonical dataset and SHA-256. The synthetic script implements Step16's 40-replication random-number stream, Ridge estimators and held-out scoring and compares all 320 scores to the archived outputs; it also writes paired synthetic 95% $t$ intervals. The frozen-export verifier checks 244 individual SHA-256 digests, the identical Step4/Step6 graph mask, pooled transition scores, all five stored DBN probability arrays, the matched DeepLOB/DBN/prior score table and all 12 gap contrasts. These are independent artifact-and-score checks, not complete refitting or an unexposed market test.

**Full market checkpoint restoration and optional rerun.** Open `Code/src/PGM_Project_Frozen.ipynb` in a Python 3 / Google Colab runtime with the originally used CausalLOB Google Drive tree mounted at `/content/drive/MyDrive/CausalLOB`. Alternatively configure `CAUSALLOB_ROOT` for cells that support it; Step2 and some older cells still use explicit Drive paths. Execute notebook cells *in their existing order*, or reuse verified matching checkpoints to avoid recomputing frozen discovery/training. Main dependencies include `tigramite==5.2.10.1`, NumPy, pandas, SciPy, scikit-learn, matplotlib, PyArrow, KaggleHub and PyTorch. The source notebook performs some of its own package installation; it is not a fully locked environment file. The original DeepLOB setup uses seed 5162026, 100-by-40 windows, batch size 256, 15 maximum epochs, early-stopping patience 3, learning rate $10^{-3}$, weight decay $10^{-5}$, and a fixed train/validation/evaluation chronology. GPU/CPU FP32 paths need not reproduce bitwise-identical training. The Step16 artificial test uses seed 5162026, 40 replications, 800/300 train/test targets and Ridge $\alpha = 1$.

**Necessary upstream artifacts.** The original notebook expects Step2 files such as `01_canonical_raw.npz`, `02_boundaries.npz`, `03_features.parquet`, `04_temporal_primary.npz` and source manifests; the Step4 frozen parent-mask NPZ; and the Step6–15 model, evaluation and DeepLOB checkpoints. Their exact paths and fail-closed hash assertions are in the supplied notebook. The included canonical NPZ has `X`, `Y`, `day` and segment flags, but does *not* have the original Step2 checkpoint schema and cannot simply be renamed to satisfy downstream assertions. If missing, Step2 can rebuild from the listed FI-2010 NoAuction Decimal Precision files via the notebook's data finder; the rest of the run must use the resulting verified checkpoints. The original Google Drive checkpoint tree is included as a byte-preserving 244-file export with its source SHA-256 manifest. Every entry was checked locally; saved model probabilities and comparison tables were also verified. The export supports restoration of the frozen checkpoint tree, but an end-to-end clean retraining of LPCMCI, Gaussian transitions and DeepLOB was *not* executed in this environment. The earlier Step16 35-file hash-parity check remains an archived assertion, distinct from the new 244-file integrity audit.

**Data and input fingerprints.** The local checks use these SHA-256 digests (shown in two readable halves):

**Canonical archive**
`e55943477e294b853ea032137ab5907a`
`53a207ff36fcbc710d7024f6d613d696`

**Frozen source notebook**
`42e3604f6a619cbc15a4372aec8a071ec`
`65ee830d5e954e6c6542f575bd657fc`

**Expected parent-mask hash (Step6)**
`eb38b17c1431c0a98d46ed1c792953`
`c932c58572799b64aee79165dfc58439da`

These are provenance identifiers, not evidence of an unseen test. The independently verified day counts, from Days 1 through 10, are 39,512; 38,397; 28,535; 37,023; 34,785; 39,152; 37,346; 55,478; 52,172; 31,937 (total 394,337). Every day contains five segment starts. The reproduction script validates each count, and the source archive exposes the original day/segment arrays.